

3.3 Mastering Memory

Have you ever felt your computer whine and groan when starting up the latest game or application? Did you notice that the program took too long to install? Did you read the system requirements when you installed the program?

After the processor, the component that plays the most crucial role in system performance is the RAM (Random Access Memory). The more you give it, the better your system runs. There is a ceiling to how much RAM you can install to actually improve system performance, but there usually is a definite improvement depending on how much you increase the RAM.

Older RAM sticks were available in the form of EDO RAM SIMMs (Pentium 1) and then transformed into a new format called SDRAM (Pentium 2 and above). Most users today have SDRAM installed in their systems, which operates at speeds of up to 133 MHz.

Previously there was just SDRAM, up to speeds of 133 MHz, which was available for computers running Pentium 3 processors. With the introduction of the Pentium 4, a new type of RAM called Double Data Rate (DDR) SDRAM came up. These RAM modules operate at a minimum of 266 MHz, which is twice the rate of the previous generation. Later, the speeds increased to 400 MHz.

If you upgrade to a non-PCIe platform, DDR 400 RAM modules are recommended. With motherboards taking advantage of dual-channel RAM, the overall system performance gets a marked boost. DDR400 256 MB RAM modules will cost you around Rs 1,700, while a DDR400 512 MB RAM stick will cost around Rs 3,300.

SDRAM for older systems are still available, but surprisingly, they cost more than DDR RAM!

PCI-Express chipsets cost more, and they use RAM of a format called DDR2, which run at a minimum of 533 MHz. These RAM